Lab 2 In Class

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```
packages <- c("tidygeocoder", "sf", "terra", "tidyverse", "tidycensus", "flextable", "tmap")

installed <- packages %in% rownames(installed.packages())

if (any(!installed)) {
  install.packages(packages[!installed])
}

lapply(packages, library, character.only = TRUE)
```

```
## Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
## terra 1.9.46
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x tidyr::extract() masks terra::extract()
## x dplyr::filter()  masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
##
## Attaching package: 'flextable'
##
##
## The following object is masked from 'package:purrr':
##
##     compose
##
##
## The following objects are masked from 'package:terra':
##
##     align, colorize, rotate, width
```

```
## [[1]]
## [1] "tidygeocoder" "stats"        "graphics"     "grDevices"    "utils"
## [6] "datasets"     "methods"      "base"
##
## [[2]]
## [1] "sf"           "tidygeocoder" "stats"        "graphics"     "grDevices"
```

```
## [6] "utils"        "datasets"     "methods"      "base"
##
## [[3]]
##  [1] "terra"        "sf"           "tidygeocoder" "stats"        "graphics"
##  [6] "grDevices"    "utils"        "datasets"     "methods"      "base"
##
## [[4]]
##  [1] "lubridate"    "forcats"      "stringr"      "dplyr"        "purrr"
##  [6] "readr"        "tidyr"        "tibble"       "ggplot2"      "tidyverse"
## [11] "terra"        "sf"           "tidygeocoder" "stats"        "graphics"
## [16] "grDevices"    "utils"        "datasets"     "methods"      "base"
##
## [[5]]
##  [1] "tidycensus"   "lubridate"    "forcats"      "stringr"      "dplyr"
##  [6] "purrr"        "readr"        "tidyr"        "tibble"       "ggplot2"
## [11] "tidyverse"    "terra"        "sf"           "tidygeocoder" "stats"
## [16] "graphics"     "grDevices"    "utils"        "datasets"     "methods"
## [21] "base"
##
## [[6]]
##  [1] "flextable"    "tidycensus"   "lubridate"    "forcats"      "stringr"
##  [6] "dplyr"        "purrr"        "readr"        "tidyr"        "tibble"
## [11] "ggplot2"      "tidyverse"    "terra"        "sf"           "tidygeocoder"
## [16] "stats"        "graphics"     "grDevices"    "utils"        "datasets"
## [21] "methods"      "base"
##
## [[7]]
##  [1] "tmap"         "flextable"    "tidycensus"   "lubridate"    "forcats"
##  [6] "stringr"      "dplyr"        "purrr"        "readr"        "tidyr"
## [11] "tibble"       "ggplot2"      "tidyverse"    "terra"        "sf"
## [16] "tidygeocoder" "stats"        "graphics"     "grDevices"    "utils"
## [21] "datasets"     "methods"      "base"
```

Geocoded marijuana dispensaries

```
download.file("https://raw.githubusercontent.com/pjames-ucdavis/SPH215/refs/heads/main/san_francisco_ac

sf_mj <- read_csv("san_francisco_active_marijuana_retailers.csv")
```

```
## Rows: 33 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (10): License Number, License Type, Business Owner, Business Structure, ...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(sf_mj)
```

```
## # A tibble: 6 x 10
##   `License Number` `License Type`          `Business Owner` `Business Structure`
##   <chr>            <chr>                   <chr>            <chr>
## 1 C10-0000614-LIC  Cannabis - Retailer Li~ Terry Muller     Limited Liability C~
## 2 C10-0000586-LIC  Cannabis - Retailer Li~ Jeremy Goodin    Corporation
```

```
## 3 C10-0000587-LIC  Cannabis - Retailer Li~ Justin Jarin     Corporation
## 4 C10-0000539-LIC  Cannabis - Retailer Li~ Ondyn Herschelle Corporation
## 5 C10-0000522-LIC  Cannabis - Retailer Li~ Ryan Hudson      Limited Liability C~
## 6 C10-0000523-LIC  Cannabis - Retailer Li~ Ryan Hudson      Limited Liability C~
## # i 6 more variables: `Premise Address` <chr>, Status <chr>,
## #   `Issue Date` <chr>, `Expiration Date` <chr>, Activities <chr>,
## #   `Adult-Use/Medicinal` <chr>
```

```
head(sf_mj$`Premise Address`)
```

```
## [1] "2165 IRVING ST san francisco, CA 94122 County: SAN FRANCISCO"
## [2] "122 10TH ST SAN FRANCISCO, CA 941032605 County: SAN FRANCISCO"
## [3] "843 Howard ST SAN FRANCISCO, CA 94103 County: SAN FRANCISCO"
## [4] "70 SECOND ST SAN FRANCISCO, CA 94105 County: SAN FRANCISCO"
## [5] "527 Howard ST San Francisco, CA 94105 County: SAN FRANCISCO"
## [6] "2414 Lombard ST San Francisco, CA 94123 County: SAN FRANCISCO"
```

```
sf_mj$Premise_Address <- gsub(" County: SAN FRANCISCO",
                              "", sf_mj$`Premise Address`)
head(sf_mj$Premise_Address)
```

```
## [1] "2165 IRVING ST san francisco, CA 94122"
## [2] "122 10TH ST SAN FRANCISCO, CA 941032605"
## [3] "843 Howard ST SAN FRANCISCO, CA 94103"
## [4] "70 SECOND ST SAN FRANCISCO, CA 94105"
## [5] "527 Howard ST San Francisco, CA 94105"
## [6] "2414 Lombard ST San Francisco, CA 94123"
```

Tidygeocoder

```
sf_mj_geo      <- geocode(sf_mj, "Premise_Address",
                          method = "osm")
```

```
## Passing 33 addresses to the Nominatim single address geocoder
```

```
## Query completed in: 34 seconds
```

Let's check our data

```
head(sf_mj_geo)
```

```
## # A tibble: 6 x 13
##   `License Number` `License Type`          `Business Owner` `Business Structure`
##   <chr>            <chr>                   <chr>            <chr>
## 1 C10-0000614-LIC  Cannabis - Retailer Li~ Terry Muller     Limited Liability C~
## 2 C10-0000586-LIC  Cannabis - Retailer Li~ Jeremy Goodin    Corporation
## 3 C10-0000587-LIC  Cannabis - Retailer Li~ Justin Jarin     Corporation
## 4 C10-0000539-LIC  Cannabis - Retailer Li~ Ondyn Herschelle Corporation
## 5 C10-0000522-LIC  Cannabis - Retailer Li~ Ryan Hudson      Limited Liability C~
## 6 C10-0000523-LIC  Cannabis - Retailer Li~ Ryan Hudson      Limited Liability C~
## # i 9 more variables: `Premise Address` <chr>, Status <chr>,
## #   `Issue Date` <chr>, `Expiration Date` <chr>, Activities <chr>,
## #   `Adult-Use/Medicinal` <chr>, Premise_Address <chr>, lat <dbl>, long <dbl>
```

```
summary(sf_mj_geo$lat)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs
##   37.71   37.75   37.78   37.77   37.78   37.80      11
```

```
summary(sf_mj_geo$long)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs
##  -122.5  -122.4  -122.4  -122.4  -122.4  -122.4      11
```

Let's try the ArcGIS database

```
sf_mj_geo_arc <- geocode(sf_mj, "Premise_Address",
                         method = "arcgis")
```

```
## Passing 33 addresses to the ArcGIS single address geocoder
```

```
## Query completed in: 13.9 seconds
```

Check our data

```
head(sf_mj_geo_arc)
```

```
## # A tibble: 6 x 13
##   `License Number` `License Type`          `Business Owner` `Business Structure`
##   <chr>            <chr>                   <chr>            <chr>
## 1 C10-0000614-LIC  Cannabis - Retailer Li~ Terry Muller     Limited Liability C~
## 2 C10-0000586-LIC  Cannabis - Retailer Li~ Jeremy Goodin    Corporation
## 3 C10-0000587-LIC  Cannabis - Retailer Li~ Justin Jarin     Corporation
## 4 C10-0000539-LIC  Cannabis - Retailer Li~ Ondyn Herschelle Corporation
## 5 C10-0000522-LIC  Cannabis - Retailer Li~ Ryan Hudson      Limited Liability C~
## 6 C10-0000523-LIC  Cannabis - Retailer Li~ Ryan Hudson      Limited Liability C~
## # i 9 more variables: `Premise Address` <chr>, Status <chr>,
## #   `Issue Date` <chr>, `Expiration Date` <chr>, Activities <chr>,
## #   `Adult-Use/Medicinal` <chr>, Premise_Address <chr>, lat <dbl>, long <dbl>
```

```
summary(sf_mj_geo_arc$lat)
```

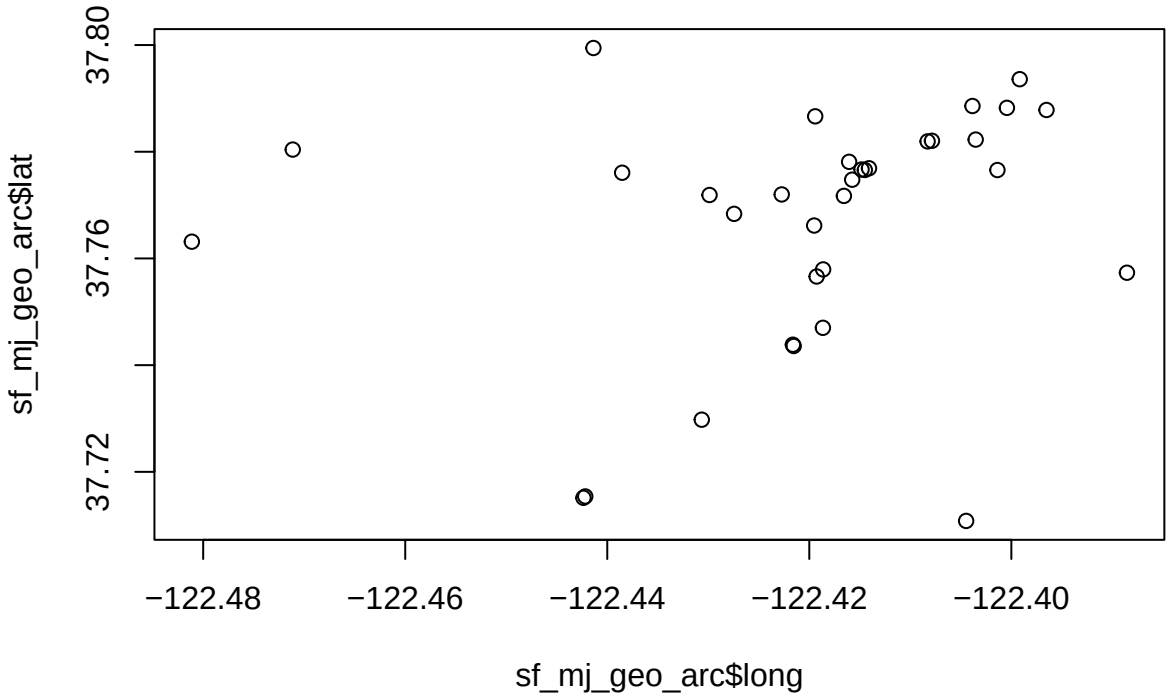
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   37.71   37.76   37.77   37.77   37.78   37.80
```

```
summary(sf_mj_geo_arc$long)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  -122.5  -122.4  -122.4  -122.4  -122.4  -122.4
```

Let's map our data!

```
plot(sf_mj_geo_arc$long, sf_mj_geo_arc$lat)
```



Working with cancer point data

Download the cancer data

```
# Download Cancer Dataset
download.file("https://raw.githubusercontent.com/pjames-ucdavis/SPH215/refs/heads/main/CAdata.rds", "CA

# Read in Cancer Dataset
cancer <- readRDS("CAdata.rds")

summary(cancer)
```

```
##       time               event              X                Y
##  Min.   : 0.004068   Min.   :0.0000   Min.   :1811375   Min.   :-241999
##  1st Qu.: 1.931247   1st Qu.:0.0000   1st Qu.:2018363   1st Qu.: -94700
##  Median : 4.749980   Median :1.0000   Median :2325084   Median : -60386
##  Mean   : 6.496130   Mean   :0.6062   Mean   :2230219   Mean   :  87591
##  3rd Qu.: 9.609031   3rd Qu.:1.0000   3rd Qu.:2380230   3rd Qu.: 318280
##  Max.   :24.997764   Max.   :1.0000   Max.   :2705633   Max.   : 770658
##       AGE          INS
##  Min.   :25.00   Mcd: 431
##  1st Qu.:53.00   Mcr:1419
##  Median :62.00   Mng:2304
##  Mean   :61.28   Oth: 526
##  3rd Qu.:71.00   Uni: 168
##  Max.   :80.00   Unk: 152
```

Make our cancer dataset spatial

```
cancer_points <- st_as_sf(cancer, coords=c("X","Y"))
```

CHECK THE CRS

```
st_crs(cancer_points)
```

```
## Coordinate Reference System: NA
```

Add coordinate reference system

```
ca_projection <- "+proj=lcc +lat_1=40 +lat_2=41.66666666666666
            +lat_0=39.33333333333334 +lon_0=-122 +x_0=2000000
            +y_0=500000.0000000002 +ellps=GRS80
            +datum=NAD83 +units=m +no_defs"

#Set CRS
cancer_points_crs <- st_set_crs(cancer_points, ca_projection)

#Look at dataset
summary(cancer_points_crs)
```

```
##       time               event             AGE         INS
##  Min.   : 0.004068   Min.   :0.0000   Min.   :25.00   Mcd: 431
##  1st Qu.: 1.931247   1st Qu.:0.0000   1st Qu.:53.00   Mcr:1419
##  Median : 4.749980   Median :1.0000   Median :62.00   Mng:2304
##  Mean   : 6.496130   Mean   :0.6062   Mean   :61.28   Oth: 526
##  3rd Qu.: 9.609031   3rd Qu.:1.0000   3rd Qu.:71.00   Uni: 168
##  Max.   :24.997764   Max.   :1.0000   Max.   :80.00   Unk: 152
##           geometry
##  POINT        :5000
##  epsg:NA      :   0
##  +proj=lcc ...:   0
##
##
##
```

Check our CRS

```
st_crs(cancer_points_crs)
```

```
## Coordinate Reference System:
##   User input: +proj=lcc +lat_1=40 +lat_2=41.66666666666666
##               +lat_0=39.33333333333334 +lon_0=-122 +x_0=2000000
##               +y_0=500000.0000000002 +ellps=GRS80
##               +datum=NAD83 +units=m +no_defs
##   wkt:
## PROJCRS["unknown",
##     BASEGEOGCRS["unknown",
##         DATUM["North American Datum 1983",
##             ELLIPSOID["GRS 1980",6378137,298.257222101,
##                 LENGTHUNIT["metre",1]],
##             ID["EPSG",6269]],
##         PRIMEM["Greenwich",0,
##             ANGLEUNIT["degree",0.0174532925199433],
##             ID["EPSG",8901]]],
##     CONVERSION["unknown",
```

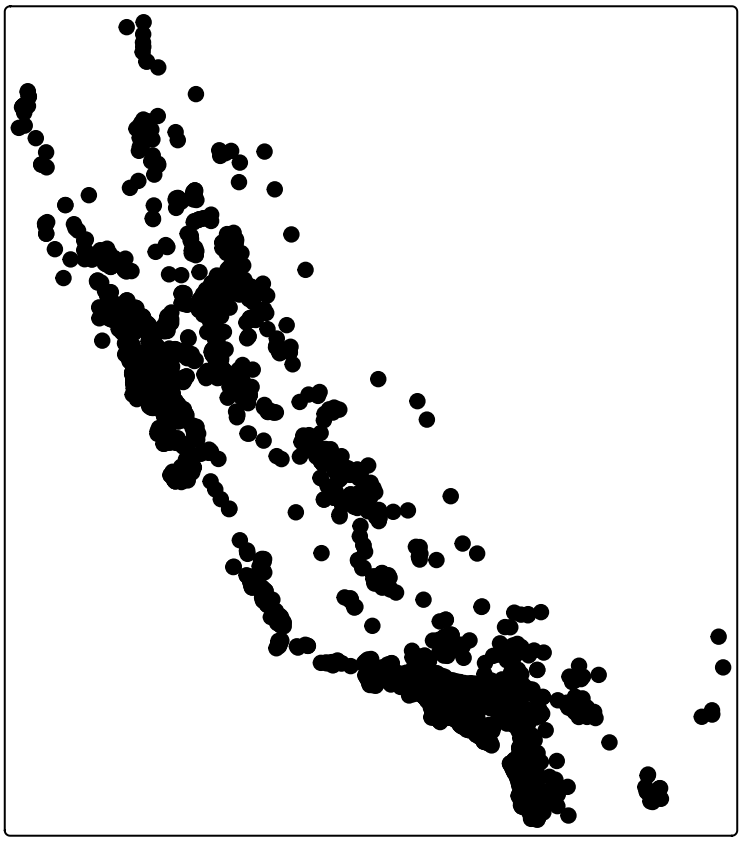
```
##         METHOD["Lambert Conic Conformal (2SP)",
##             ID["EPSG",9802]],
##         PARAMETER["Latitude of false origin",39.3333333333333,
##             ANGLEUNIT["degree",0.0174532925199433],
##             ID["EPSG",8821]],
##         PARAMETER["Longitude of false origin",-122,
##             ANGLEUNIT["degree",0.0174532925199433],
##             ID["EPSG",8822]],
##         PARAMETER["Latitude of 1st standard parallel",40,
##             ANGLEUNIT["degree",0.0174532925199433],
##             ID["EPSG",8823]],
##         PARAMETER["Latitude of 2nd standard parallel",41.6666666666667,
##             ANGLEUNIT["degree",0.0174532925199433],
##             ID["EPSG",8824]],
##         PARAMETER["Easting at false origin",2000000,
##             LENGTHUNIT["metre",1],
##             ID["EPSG",8826]],
##         PARAMETER["Northing at false origin",500000,
##             LENGTHUNIT["metre",1],
##             ID["EPSG",8827]]],
##     CS[Cartesian,2],
##         AXIS["(E)",east,
##             ORDER[1],
##             LENGTHUNIT["metre",1,
##                 ID["EPSG",9001]]],
##         AXIS["(N)",north,
##             ORDER[2],
##             LENGTHUNIT["metre",1,
##                 ID["EPSG",9001]]]]
```

Start working with tmap

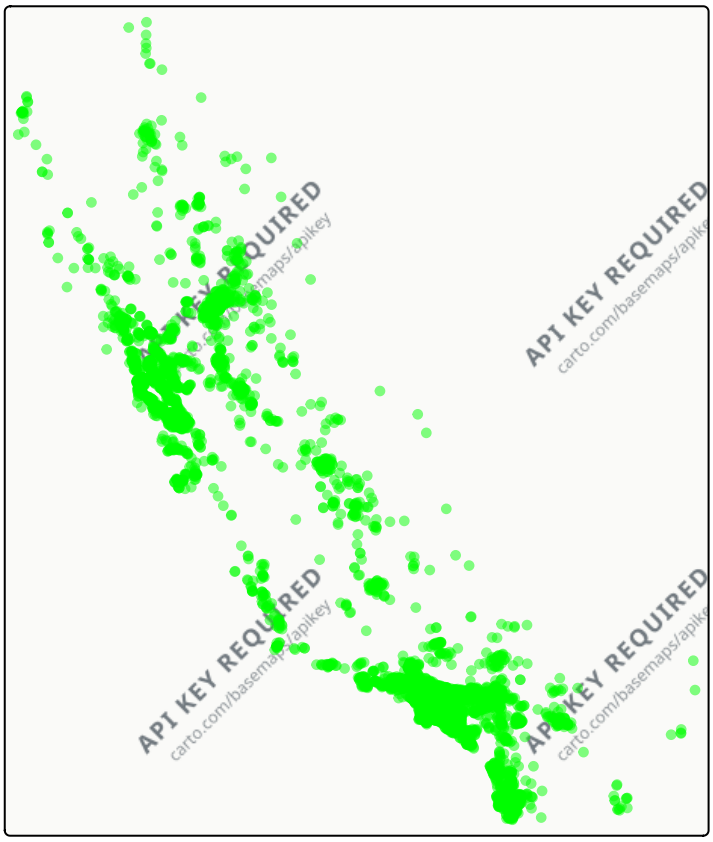
```
tmap_mode("plot")
```

```
## i tmap modes "plot" - "view"
## i toggle with `tmap::ttm()`
```

```
cancer_map <- tm_shape(cancer_points_crs) + tm_dots(size=0.5)
cancer_map
```

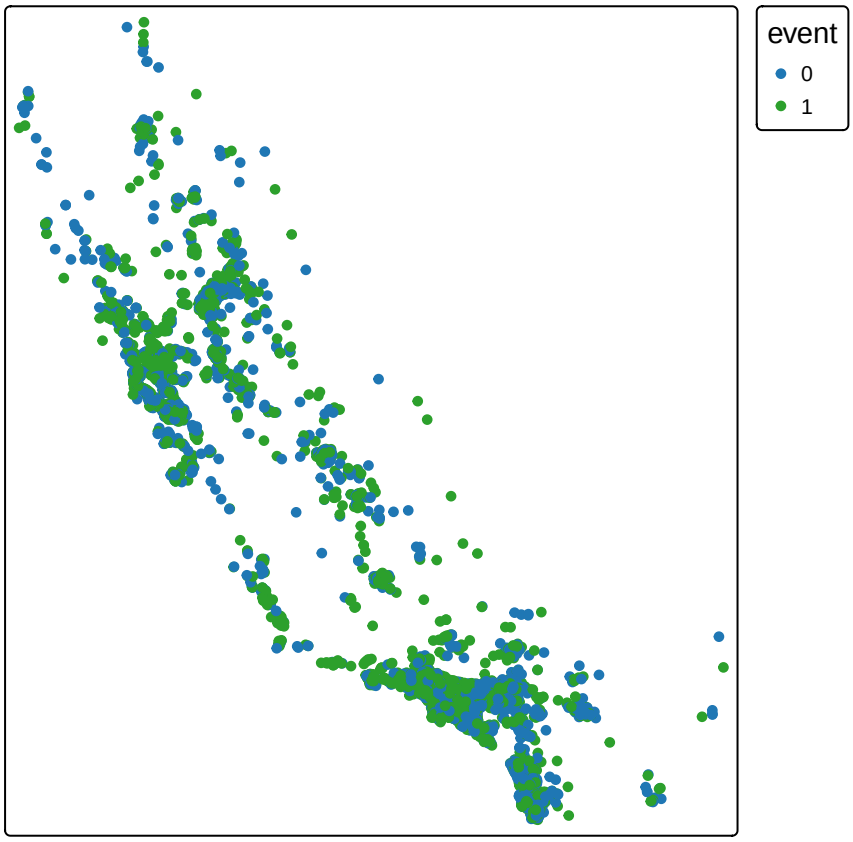


```
cancer_map_small = tm_shape(cancer_points_crs) +
  tm_dots(fill = "green", size = 0.3, fill_alpha = 0.5) +
  tm_basemap("CartoDB.Positron")
cancer_map_small
```



visualize by event status

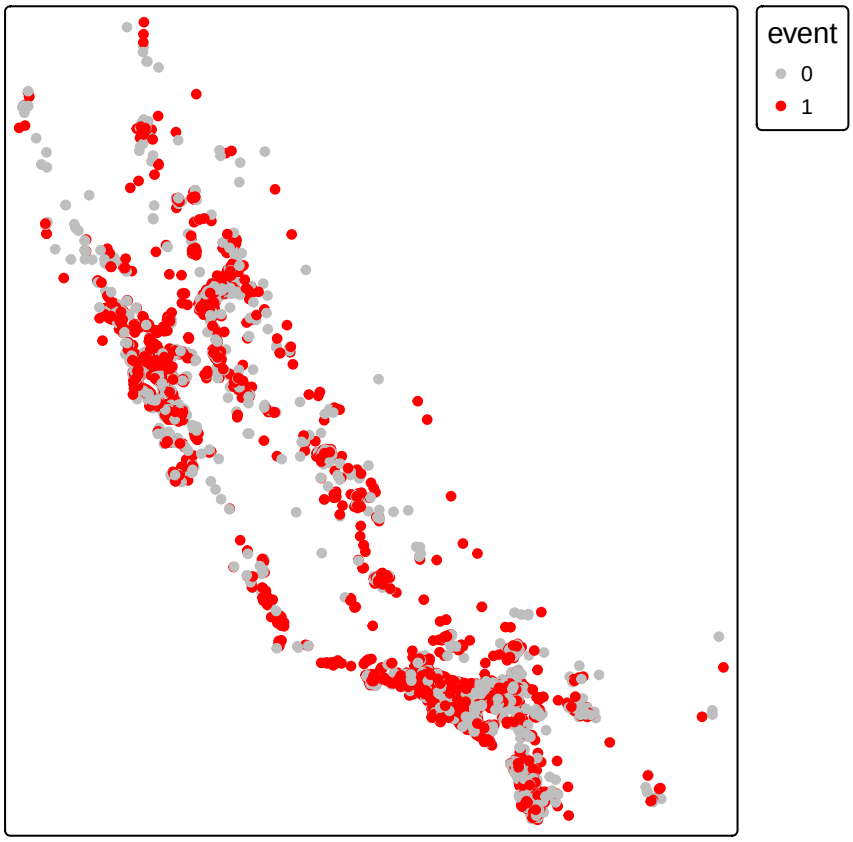
```
cancer_map_events = tm_shape(cancer_points_crs) +
  tm_dots(size=0.3, fill="event", fill.scale=tm_scale_categorical())
cancer_map_events
```



change color scheme

```
cancer_map_events_rg = tm_shape(cancer_points_crs) +
  tm_dots(fill = "event", fill.scale=tm_scale_categorical(values=c("0"="gray","1"="red")))
cancer_map_events_rg
```

```
## Multiple palettes called "gray" found: "matplotlib.gray", "tableau.gray", "ocean.gray", "gmt.gray". T
```



Import Census data

```
acs2023 <- load_variables(2023, "acs5", cache = TRUE)
```

```
## Warning: `cache` is deprecated and ignored. tidycensus no longer writes
## variable metadata to a local cache.
```

```
View(acs2023)
```

```
ca <- get_acs(geography = "county",
              year = 2023,
              variables = c(tpopr = "B03002_001",
                            nhwhite = "B03002_003", nhblk = "B03002_004",
                            nhasn = "B03002_006", hisp = "B03002_012"),
              state = "CA",
              survey = "acs5",
              output = "wide")
```

```
## Getting data from the 2019-2023 5-year ACS
```

```
glimpse(ca)
```

```
## Rows: 58
## Columns: 12
## $ GEOID    <chr> "06001", "06003", "06005", "06007", "06009", "06011", "06013"~
## $ NAME     <chr> "Alameda County, California", "Alpine County, California", "A~
## $ tpoprE   <dbl> 1651949, 1695, 41029, 209470, 45995, 21895, 1161458, 27293, 1~
## $ tpoprM   <dbl> NA, 234, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
```

```
## $ nhwhiteE <dbl> 466445, 993, 30234, 139527, 35599, 6869, 455961, 16668, 14254~
## $ nhwhiteM <dbl> 1170, 215, 341, 767, 318, 133, 1843, 199, 589, 979, 75, 554, ~
## $ nhblkE   <dbl> 159042, 0, 781, 3550, 529, 311, 94864, 805, 1522, 42060, 158,~
## $ nhblkM   <dbl> 1736, 14, 143, 393, 140, 44, 1594, 123, 232, 1334, 126, 271, ~
## $ nhasnE   <dbl> 528377, 8, 587, 11010, 1066, 101, 212373, 821, 9640, 108809, ~
## $ nhasnM   <dbl> 2269, 8, 129, 497, 238, 159, 2008, 264, 416, 1332, 182, 403, ~
## $ hispE    <dbl> 385245, 249, 6361, 40829, 6403, 13639, 316799, 5350, 27230, 5~
## $ hispM    <dbl> NA, 115, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
```

```
head(ca)
```

```
## # A tibble: 6 x 12
##   GEOID NAME  tpoprE tpoprM nhwhiteE nhwhiteM nhblkE nhblkM nhasnE nhasnM  hispE
##   <chr> <chr>  <dbl>  <dbl>    <dbl>    <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
## 1 06001 Alam~ 1.65e6     NA   466445     1170 159042   1736 528377   2269 385245
## 2 06003 Alpi~ 1.69e3    234      993      215      0     14      8      8    249
## 3 06005 Amad~ 4.10e4     NA    30234      341    781    143    587    129   6361
## 4 06007 Butt~ 2.09e5     NA   139527      767   3550    393  11010    497  40829
## 5 06009 Cala~ 4.60e4     NA    35599      318    529    140   1066    238   6403
## 6 06011 Colu~ 2.19e4     NA     6869      133    311     44    101    159  13639
## # i 1 more variable: hispM <dbl>
```

Bring in external datasets

```
setwd("/Users/pjames1/Dropbox/UC Davis Folders/SPH 215 GIS and Public Health/Github_Website/SPH215/")
ca.pm <- read_csv("PolicyMap Data 2025-03-27 192555 UTC.csv", skip = 1)
```

```
## Rows: 58 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (8): GeoID_Description, GeoID_Name, SitsinState, GeoID, GeoID_Formatted,...
## dbl (2): mhhinc, GeoVintage
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
glimpse(ca.pm)
```

```
## Rows: 58
## Columns: 10
## $ GeoID_Description <chr> "County", "County", "County", "County", "County", "C~
## $ GeoID_Name        <chr> "Alameda", "Alpine", "Amador", "Butte", "Calaveras",~
## $ SitsinState       <chr> "CA", "CA", "CA", "CA", "CA", "CA", "CA", "CA", "CA"~
## $ GeoID             <chr> "06001", "06003", "06005", "06007", "06009", "06011"~
## $ GeoID_Formatted   <chr> "=\"06001\"", "=\"06003\"", "=\"06005\"", "=\"06007\~
## $ mhhinc            <dbl> 126240, 110781, 81526, 68574, 79877, 75149, 125727, ~
## $ TimeFrame         <chr> "2019-2023", "2019-2023", "2019-2023", "2019-2023", ~
## $ GeoVintage        <dbl> 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022~
## $ Source            <chr> "Census", "Census", "Census", "Census", "Census", "C~
## $ Location          <chr> "California (State)", "California (State)", "Califor~
```

```
cacounty <- ca %>%
      left_join(ca.pm, by = c("GEOID" = "GeoID")) %>%
      mutate(pwhite = nhwhiteE/tpoprE, pasian = nhasnE/tpoprE,
             pblack = nhblkE/tpoprE, phisp = hispE/tpoprE,
            mhisp = case_when(phisp > 0.5 ~ "Majority",
                              TRUE ~ "Not Majority")) %>%
```

```
      rename(County = GeoID_Name) %>%
      select(GEOID, County, pwhite, pasian, pblack, phisp, mhisp, mhhinc)
glimpse(cacounty)
```

```
## Rows: 58
## Columns: 8
## $ GEOID  <chr> "06001", "06003", "06005", "06007", "06009", "06011", "06013", ~
## $ County <chr> "Alameda", "Alpine", "Amador", "Butte", "Calaveras", "Colusa", ~
## $ pwhite <dbl> 0.28236041, 0.58584071, 0.73689342, 0.66609538, 0.77397543, 0.3~
## $ pasian <dbl> 0.319850673, 0.004719764, 0.014306954, 0.052561226, 0.023176432~
## $ pblack <dbl> 0.0962753693, 0.0000000000, 0.0190353165, 0.0169475343, 0.01150~
## $ phisp  <dbl> 0.2332064, 0.1469027, 0.1550367, 0.1949157, 0.1392108, 0.622927~
## $ mhisp  <chr> "Not Majority", "Not Majority", "Not Majority", "Not Majority",~
## $ mhhinc <dbl> 126240, 110781, 81526, 68574, 79877, 75149, 125727, 66780, 1061~
```

```
cacounty %>%
  summarize(Mean = mean(mhhinc))
```

```
## # A tibble: 1 x 1
##     Mean
##    <dbl>
## 1 87001.
```

```
cacounty <- cacounty %>%
    mutate(region = case_when(County == "Sonoma" | County == "Napa" |
                              County == "Solano" | County == "Marin" |
                              County == "Contra Costa" | County == "San Francisco" |
                              County == "San Mateo" | County == "Alameda" |
                              County == "Santa Clara" ~ "Bay Area",
                              County == "Imperial" | County == "Los Angeles" |
                              County == "Orange" | County == "Riverside" |
                              County == "San Diego" | County == "San Bernardino" |
                              County == "Ventura" ~ "Southern California",
                              County == "Fresno" | County == "Madera" |
                              County == "Mariposa" | County == "Merced" |
                              County == "Tulare" |
                              County == "Kings" ~ "Central Valley",
                              County == "Alpine" | County == "Colusa" |
                              County == "El Dorado" | County == "Glenn" |
                              County == "Placer" | County == "Sacramento" |
                              County == "Sutter" | County == "Yolo" |
                              County == "Yuba" ~ "Capital Region",
                              TRUE ~ "Rest"))
```

```
cacounty %>%
  group_by(region) %>%
  summarize(Mean = mean(mhhinc))
```

```
## # A tibble: 5 x 2
##   region                Mean
##   <chr>                <dbl>
## 1 Bay Area           129297.
## 2 Capital Region      89288.
## 3 Central Valley      69265.
## 4 Rest                74959.
## 5 Southern California 91332.
```